

Did the new pricing actually work?

The congestion charge works in one narrow place and is getting credit for a lift it mostly did not cause; scaling it spends the one structural edge NY has. The real leaks are elsewhere: a driver cancelling over an unpaid deadhead, and a rider who gets no quote and silently leaves.

Part 1 — Frame the problem · Driver cancellation leads; the charge is a two-signed force, not a general lever. Five-test causation ladder.

Part 2 — Did the pricing work? · Real win on Auto Priority at peak ($r = +0.80$). The +14.9% headline is mostly arithmetic.

Part 3 — People & behaviour · The driver optimises the hour, not the trip. An information fix beats a price fix. The rider who gets nothing.

Part 4 — The constraint · Supply-shaping: completion without subsidy, using data NY already owns.

Part 5 — Stress test · Four critiques of the AI analysis. How a correct calculation produced a wrong recommendation.

Close — Summary for ops + exec · Do / stop-hold / one number to watch.

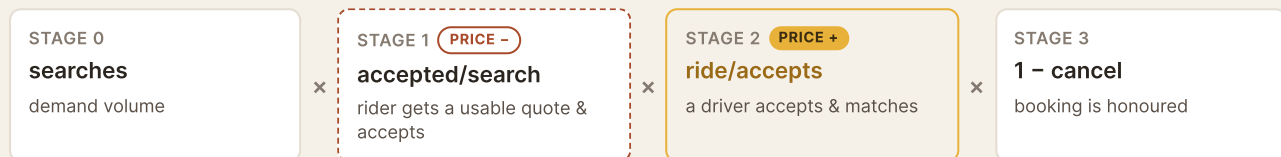
Analysis by Yagnesh L Pazhanyappan · June 2026

PART 1

Frame the problem

Completed rides are the last survivor of four multiplied stages. Nothing moves the number without moving one of those terms, which means the congestion charge is not a general lever; it is a two-signed force living in two specific spots: pulling a driver in at Stage 2, and potentially pushing a price-sensitive rider out at Stage 1.

completed rides =



The one link the congestion charge is built to move is to pull a driver into Stage 2. The same charge pushes the other way at Stage 1, where a higher fare can send a price-sensitive rider to a cheaper competitor. One lever, two signs, two stages.

Ranked hypotheses

H1 — Stage 3: driver cancellation. [Most likely] The driver optimises his hour, not this trip. A ₹67 fare with a 1.5 km pickup earns him around ₹11/km effective after deadhead, not the meter's ₹15. He accepted on the only number the offer card showed him; the cancellation is the first moment the product let him see the real trip. Multi-homing sharpens it: the only rival offer visible while he holds your booking comes from Uber or Ola, and NY cannot outbid a subsidized competitor on a per-trip basis.

H2 — Stage 0: demand volume. [High] Sets the ceiling above everything downstream. Moves with competitor subsidies, the calendar, and paydays; mostly outside our control. Product has fewer real levers here than on cancellation.

H3 — Stage 2: getting the driver to accept the ride. [Medium; the lever under test] The congestion bump lifts the fare on the trips the old meter underprices at peak. Works at the exact margin of a skipped trip; disappears in the average. A ₹8 nudge on a ₹75 fare.

H4 — Stage 1: the rider's side. [Medium; the invisible one] Two failures: the rider who gets no ride at all (a supply-thinness problem, invisible in the data), and the rider who gets priced off our platform. Same dial, opposite sign.

H5 — External forces: weather, calendar, competitor spend. [Low base rate, high variance] The most likely candidate to be doing the work the pricing gets credit for, given the timing coincidence.

How I'd tell cause from coincidence

The rollout was never a clean A/B: version assignment was observational and outcome-dependent, with winning versions ramped toward 100% fast. A slice must clear five tests in order before I credit the pricing:

1. **Price moved.** Rebuild applied multiplier from `congestion_sum / congestion_n`. If it never rose, stop here.
2. **Like-for-like.** Same tier, distance band, and hour against the identical slice a week prior, so confounds are held still rather than averaged away.
3. **Clean comparison window.** v250 vs v252 inside the same peak hours: time of day matched, pricing logic the only variable left.

4. **Dose-response + placebo.** Completion rises more where surge rose more, and stays flat where price never moved.
5. **Denominator check + both sides.** Track raw completed rides, not just the ratio. Compare NY fare to competitors at Stage 1 (defection risk) and Stage 3 (poach risk) simultaneously.

I ranked by which stage holds the binding constraint, weighing how big the leak is against how directly product can move it. H1 leads because Stage 3 is the biggest and the last leak, and a driver's earnings-per-hour logic predicts it with no data at all. H2 follows because Stage 0 sets the ceiling, even though we have weaker levers on it. H3, the lever under test, lands in the middle honestly: real, but funded only by a small rider bump under zero-commission, and active at a narrow margin rather than across the board. H4 sits next to it because it carries the same price story's other sign and the invisible missing-quote loss. H5 ranks last on base rate but stays in the frame for its variance.

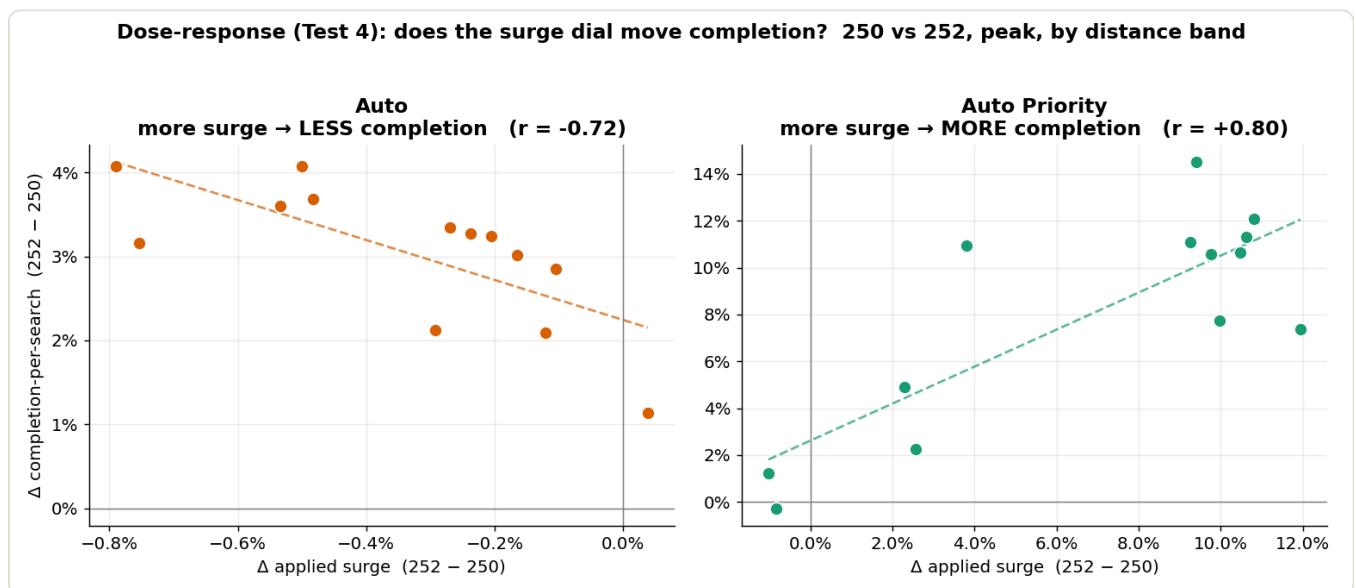
The belief I most expect to be wrong is H3 read as a broad win, which is exactly the belief leadership is leaning on. I expect the gains to be narrow: peak, short, busy-area trips, and mostly in Auto Priority, whose riders care least about an extra ₹15. I expect the rupee effect to be small, and I expect some of it to leak straight back out at Stage 1 as riders with the higher fare are pushed to a cheaper competitor. The multi-homing angle makes me more sure, not less. If the biggest leak is cancellation, and a real chunk of cancellation is drivers getting poached by a subsidized competitor, then pricing cannot fix that part at all, because NY cannot outbid a subsidy. My working bet is that the two-week tick up is mostly H5 and H2, a recovery and a shift in the demand mix, credited to H3 because the timing happened to line up.

PART 2

Did the pricing actually work?

The +14.9% headline is real; I can reproduce it to the decimal. It is also mostly not the pricing. Two findings survive five causation tests, and they point in opposite directions.

The chart that frames both: the dose-response, the 250-vs-252 peak comparison split by tier. The two tiers do not just differ in degree; they have opposite signs.

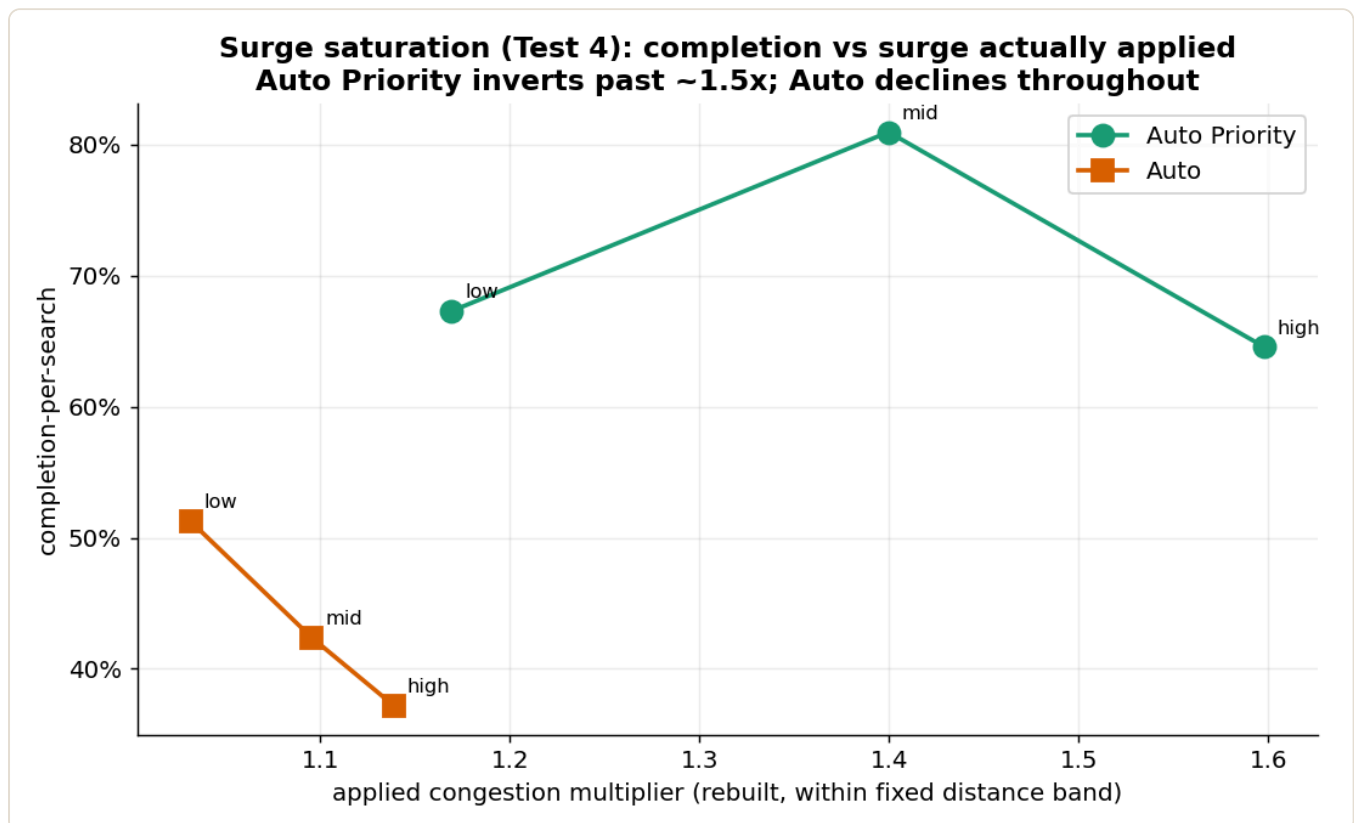


Finding 1 — Auto Priority at peak: the pricing genuinely works. Small, specific, and real.

- v250 vs v252 inside the same peak hours, same tier, same distance bands. Only the pricing logic varies.

- Textbook dose-response: the bands where 252 added the most surge are exactly where completion rose most. At 4-5 km: multiplier 1.41 → 1.52, completion 0.770 → 0.890. At 9-10 km: similar surge bump bought a +0.145 jump.
- Bands where 252 barely changed the surge (12-16 km and beyond) show no lift at all.
- Correlation between how much surge rose and how much completion rose: $r = +0.80$, stable when weighted by sample size. No thin cells steering the line: lightest band ~1,100 searches, most between 2,000 and 5,000.
- Competitor gate cleared: NY was 5-33% cheaper than rivals across every band throughout, and drifted cheaper still over the rollout (from -10% to -18% on Auto Priority peak). No rider defection explains this lift.

The ceiling. Completion traces an inverted-U with surge: it climbs to around 1.4x, then falls above ~1.6x. The 250-to-252 move sat in the productive zone. The dial has room but it is narrow.



Finding 2 — The headline +14.9% is mostly not the pricing. Three independent tests say so.

Test 1: Dose-response runs backwards on mass Auto. In the same 250-vs-252 peak comparison, 252 completes better in every band but does so with lower surge and a lower average fare (₹152 vs ₹154). When I bin by the surge actually applied within a fixed band, more surge tracks less completion: at 4-5 km, completion falls from 0.518 in the low-surge bin to 0.372 in the high-surge bin. The surge is reading market stress, not relieving it.

Test 2: The placebo lights up. A charge that only fires at peak cannot move hours where it never fired. Yet Auto completion rose just as much at night (0.335 → 0.437) and off-peak (0.430 → 0.522). Two things actually moved: a version swap buried in the rollout (v211 → v236/v250, which simply clears better at identical surge, 0.596 vs 0.558), and a supply expansion that predated the congestion logic.

Test 3: Count, not ratio. Auto searches dropped from 181k to 168k per day; completion-per-search climbed partly because fewer people were asking. The raw count of finished Auto rides began recovering before the charge shipped: 82,900/day in late May to 88,900 in early June, with the post-rollout window catching only the

tail on a shrinking base.

PART 3

Get inside their heads

3.1 — The driver who accepts a ₹67 trip, then cancels

He is not optimising the trip; he is optimising the hour. A ₹67 fare with a 1.5 km pickup means driving six kilometres to get paid for four and a half, dropping his effective rate from ₹15/km to around ₹11 before fuel. He accepted on the only number the offer card showed him, because hesitating risked losing the ping. The cancellation is the first moment the product let him see the real trip. That is a UX failure wearing the costume of a behaviour problem.

He is also missing three things that would settle the decision: the destination (does this drop me somewhere dead?), whether a better ping is seconds away (NY dispatches one at a time; the only rival offer visible mid-hold comes from Uber or Ola, who cannot tell he is busy), and what this corner looks like in twenty minutes. He is making an earnings-per-hour call with almost none of the inputs that decision needs.

Would a congestion bump change his mind? Barely. On the mass tier the surge moved six rupees on a ₹150 fare. Six rupees does not redeem a bad trip, and where it did rise on Auto, completion fell, because the charge fires hardest exactly when the market is most stretched. The money is aimed at the wrong moment; its label and timing are wrong, not just its size.

3.2 — Experiment: the legible deadhead

Hypothesis. Most driver cancellations on short trips with long pickups are an information problem, not a pay problem. The driver accepts blind to the deadhead, then cancels the instant the product shows it to him.

Intervention. Two linked changes on peak slices where a congestion charge already applies (because that is where the rupee to rebrand actually exists):

- The offer card leads with pickup distance and a net-of-deadhead number: "₹67, minus ~₹18 deadhead, ~₹49 net," instead of the bare fare.
- The congestion uplift is renamed and surfaced as "deadhead pay: +₹X for your 1.5 km pickup," converting an abstract area tax into named compensation for the exact friction he cancels over.

The rider sees the same total fare, presented as "busy-area pickup support." Same rupee, one coherent story to both sides.

Prerequisite. Audit the live driver app first. The entire experiment rests on one premise: the offer card today shows fare but not deadhead-adjusted economics. If proximity matching or pre-accept transparency is already shipped, this design is moot or must be re-scoped.

Primary metric. Driver-cancellation rate on the treated slice vs the same slice the prior week. Cleaner where the funnel allows: honour rate (completed / accepted bookings).

Guardrail metrics:

- Acceptance rate and no-quote share (a cancellation pushed upstream is not fixed).
- Completion-per-search and competitor fare gap on the rider side (the fare did not rise; if these move, something leaked).
- Auto and Auto Priority read separately, never blended.

Gaming surfaces. Accept-then-stall (driver accepts, sits still, forces rider to cancel): watch

cancelled_by_user share, time-to-cancellation distribution, and frozen-driver-location patterns after accept. Deadhead farming (hovering past a distance threshold to trigger the bonus): watch pickup-distance distribution for unnatural clustering at the fee boundary.

The seam I will not hide. The congestion charge is area-based; the deadhead is trip-based. They only fund each other where they coincide: at peak in busy zones. Off-peak trips with long pickups have no rupee to rebrand, which is exactly why I scope the first test to peak. If it works there, the open question is whether off-peak deadhead is worth funding with an explicit rider pickup fee, which reopens the Stage 1 defection risk and would need its own test.

3.3 — The rider who searches and gets nothing

Her moment is short and quiet, which is why it is dangerous. She searches, waits, and either gets no quote or one that is too expensive. She closes the app. Namma Yatri can see the search did not become a ride, but not why: a no-quote, a price-sensitive rejection, and idle curiosity all collapse into the same shortfall, and the app she opened next is invisible.

A no-quote is worse than a cancellation. A cancellation leaves a fingerprint; a no-quote removes a customer and a sliver of network liquidity at once and tells you nothing. On a platform where both sides multi-home, that sliver is gone before you can count it: the driver it would have brought online logs into Uber instead, making the next rider's search harder, which drives the next driver away. The quiet exits are how a marketplace unwinds, one at a time, each making the next more likely.

3.4 — From my own life

I moved to a new city for work this year, and the job keeps me on the road about a fifth of the time, so the last twelve months have made me a heavy and slightly obsessive user: north of 200 rides across autos, cabs, and bikes, mostly in Mumbai, Bangalore, and Chennai, with a handful in tier-1.5 cities like Coimbatore and Visakhapatnam. Ride that much in that many places and you stop seeing one market; you start seeing several. A few of the patterns line up uncomfortably well with what the cube said, and one cuts against an assumption baked into this whole brief.

That assumption is that pricing is a single lever that behaves the same way everywhere, and it is not. Multi-homing is universal; every driver I have spoken to runs several apps at once. But what a driver does with a ride he has accepted changes from city to city. In some places asking for a little over the meter is a candid part of the trip, so he accepts and then negotiates; in others the app's own matching does the price discovery, and a driver who finds the fare not worth it simply never quotes, instead of accepting and squeezing. That is not a small UX detail. It means the job a user (rider/driver) hires the app to do is genuinely different in different cities: in one it is a discovery platform that is suggesting price, in another its a full marketplace where price is fixed when match happens.

The second pattern is about the trips a fare cannot rescue. Weather and the traffic on the specific route impact acceptance far more than the headline number does; I have watched drivers wave off a well-paid trip because the route itself was misery, and some routes are not worth taking even at fifty percent more. That is Part 2's saturation curve seen from the front seat: past a point the driver is pricing the road, not the meter, and another rupee buys nothing. It is also why a one-way run into a deadzone in off-peak hours is a last resort, taken to stop the bleed of sitting idle rather than because it pays; he is managing loss aversion on his earnings per hour, not chasing the trip.

Fill rates are visibly better on the zero-commission apps, and what decides which app a driver actually favours is rarely the fare; it is the non functional requirements like, how fast the online payment settles and whether a promised promotion really lands. Another qualm is that the pricing is slow to react to cost, taking more than a couple of weeks after fuel price moves to climb back to a driver's real breakeven and the margin he had before.

PART 4

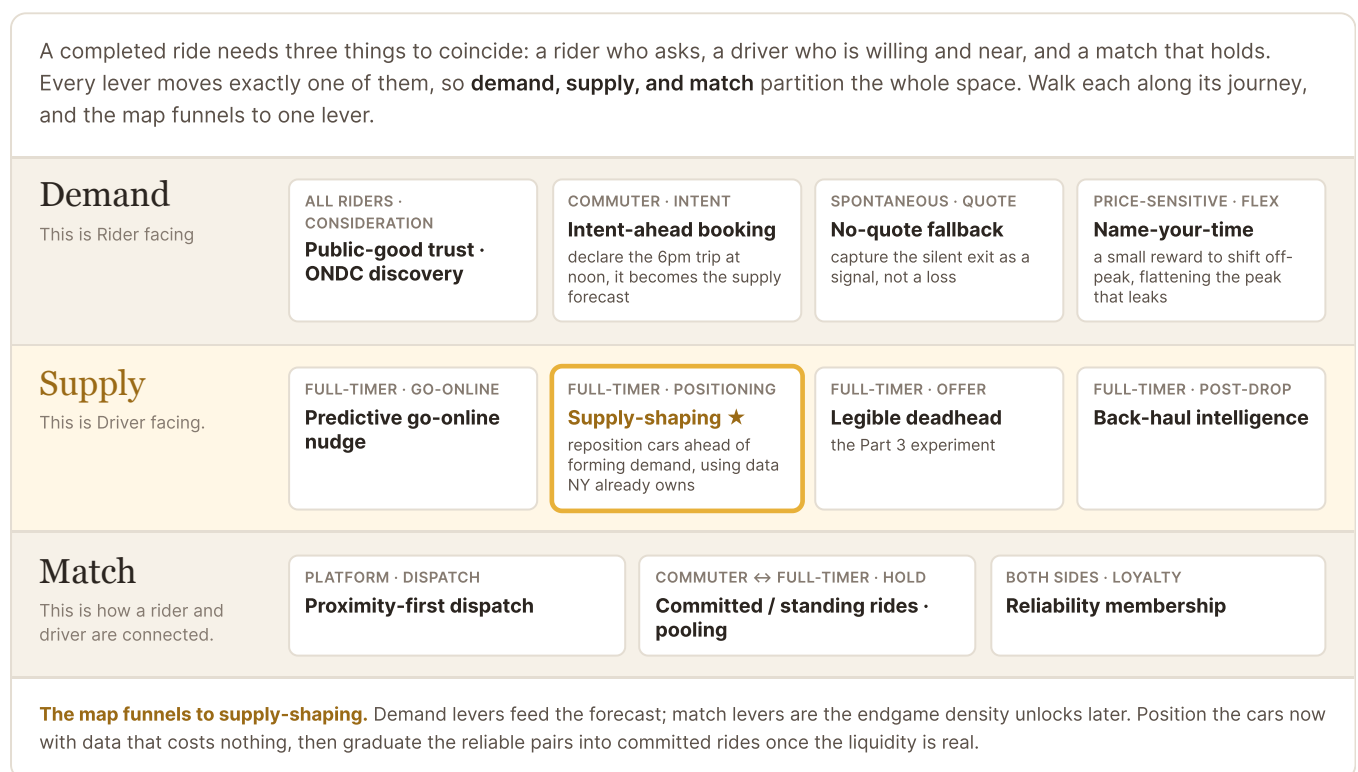
The constraint

Why raising the fare is the wrong lever

Under zero-commission, every rupee of the congestion charge comes from the rider at one-to-one. It saturates near 1.5x. And it aims at the wrong leak: the driver cancelling over an unpaid deadhead and the rider getting no usable quote are not problems a price change fixes. NY has nominal headroom, about 17% below the cheaper rival on mass Auto at peak, but spending that cushion is the wrong move because the cushion is the moat. NY's durable advantage over a subsidized rival is being the cheapest honest option built on the lowest cost to serve; you do not defend that by becoming less cheap. The right move is to stop buying completion with money NY does not have and start buying it with the asset it does own: data and prediction.

The option space

A completed ride needs exactly three things to coincide: a rider who asks, a driver who is willing and near, and a match that holds. Nothing else matters. Every conceivable lever moves exactly one of these, which makes **demand, supply, and match** a complete, non-overlapping segments of the whole solution space.



Where the map converges. At evening peak, the moment the funnel leaks most, demand is not the scarce input; most searches are happening right then. The binding constraint is supply not being where they ask. Demand-side levers feed the forecast but do not close the gap. Match-structure fixes like committed rides are the most complete answer, but they rest on a promise of dense, reliable supply that does not yet exist. That leaves the supply column, and within it positioning, as the one cell where the binding constraint, the data we actually have, and the zero-rupee limit all intersect.

Competitor read and the defection risk of pricing higher

Comparing within distance band matters here; the competitor scrape skews long (~16 km median) and a pooled average is distance-confounded.

- **Mass Auto at peak:** NY sits ~17% below the cheaper rival, deepest on short trips (28% under at 1-2 km). Real headroom, but spending it is the wrong call; the cushion is the moat.
- **Auto Priority peak at 3-6 km:** NY already sits above the cheaper rival by 3-12%. That is the same slice Part 2 credited as the one real pricing win. Raise congestion there and you cross the market in the exact bands that were carrying the result, with a subsidized rival one tap away.

The defended lever: supply-shaping

The idea. Position drivers ahead of predictable demand. A driver idling in a thinning zone gets a concrete, self-interested nudge toward a forming hotspot in his own terms: "Move 1.2 km to Indiranagar; 18 searches forming there in 15 minutes, average pickup 0.4 km against your 1.5 km here." Not a price change, not a subsidy; the platform spending its information to put the right car near the right rider before either has to settle for a bad match.

The honest prerequisite. This lever does not run on the current cube since geography is the missing input. a repositioning nudge is geography dependant. The first step is the instrumentation: log pickup distance, tag supply and demand to zones. That signal exists in the operational system today, because NY already runs this dispatch; it simply never reached the analysis cube. The gap is instrumentation and access, not a missing capability.

Why demand is forecastable enough. Weekday search volume by hour is highly regular (median hourly CV ~0.26). The evening peak's timing and location recur every weekday. Completion-per-search hits its worst of the day at 6-7 pm (0.37 and 0.35), with cancellations at their daily high. The biggest, most forecastable hole in the funnel announces itself on a schedule.

Win-win-win. The driver wins first: shorter pickups mean less unpaid deadhead, which lifts his rupees-per-hour, the exact number Part 3 said the bad trip was wrecking. The rider wins because a pre-positioned car converts her search into a quote. NY wins because thicker liquidity at the predictable peak is completion without subsidy, and every pickup it shortens is network capital it keeps rather than leaks.

Zero marginal cost at scale. The prediction runs on data NY already holds and compute it already owns. A subsidized rival grows completion by spending money it does not make; NY grows it by spending intelligence it already has.

The second-order risk. A naive broadcast that tells every idle driver about the same hotspot creates a glut where they land and a new shortage where they left. The fix is allocation, not announcement: send roughly as many cars as the forming demand can absorb, stagger the nudges, and read supply alongside demand. A nudge that proves wrong a few times gets ignored forever; the prediction has to be good before it ships.

Why this one first. Supply-shaping thickens the base that makes everything downstream credible.

Committed rides are the more complete fix, but they require guaranteed supply density that does not yet exist. Build the density with information that costs nothing, then graduate reliable pairs into standing rides once the liquidity is real. NY does not have the subsidy to buy its way to that density; it has the data to route there.

PART 5

Stress-test the AI analysis

The analysis computed one number correctly and then asked it to carry a decision it cannot bear. I reproduced the +14.9% headline exactly, to the decimal, so there is no hiding behind a computation dispute. What follows is why a correct calculation produces a wrong recommendation.

Critique 1 — The headline is a confound wearing a causal claim

The method compares all Bangalore auto transactions before vs after the rollout. The versions were never cleanly assigned: v250/252 run mainly at peak; v236 runs mainly off-peak and at night. A before-and-after is peak vs off-peak, not pricing vs no-pricing.

Three checks dismantle the +14.9%:

- The lift appears equally at night (Auto: 0.335 → 0.437) and off-peak (0.430 → 0.522), where the charge never fired. A lever that only acts at peak cannot move hours it never touched.
- The denominator fell: Auto searches dropped from 181k to 168k per day. Completion-per-search climbed partly because fewer people were asking.
- Raw Auto completed rides began recovering before the charge shipped: 82,900/day in late May to 88,900 in early June.

Ship implication: the number anchoring the recommendation is not measuring the pricing. The one clean comparison the rollout left behind (v250 vs v252 at peak) shows a narrower, tier-specific effect, which is the opposite of a green light for a blanket rollout.

Critique 2 — The asserted mechanism is absent, and the analysis contradicts itself

Finding 5 of the analysis states the causal story: "congestion pricing increases driver supply, which is why completion improved." On the mass Auto tier that mechanism is physically absent. The applied multiplier moved from 1.062 to 1.068, about ₹6 on a ₹150 fare. Finding 2 then celebrates that "average fare barely moved (+1.5%)." These two findings cannot both be true: a fare that barely moved cannot have raised supply. The analysis also predicts the effect "should help most on longer trips," which the data rejects: the only genuine dose-response concentrates on short and mid bands, while the long bands it predicted show no lift at all.

Ship implication: a 100% Auto rollout bets on a causal channel the data denies. Scaling the surge on the mass tier scales a placebo, and the real leak, cancellation over the deadhead, stays exactly where it is.

Critique 3 — "Auto Priority loves it" mistakes a ratio for rides

The analysis reads 0.77–0.85 completion-per-search on Auto Priority as proof "riders love congestion pricing." Auto Priority's completed rides per day actually fell after the rollout, from ~16,900 to ~15,900, because its search denominator collapsed from 25,400 to 20,800. The tier the analysis leans on hardest delivered fewer finished rides. Completion-per-search is also a marketplace clearing rate, not a satisfaction signal; it structurally cannot see the riders who got no usable quote and silently left, who are the ones a price increase would push away first.

Ship implication: the single strongest piece of evidence in the analysis points the wrong way when you count rides instead of ratios. Crowning the premium tier the winner while its absolute volume shrinks is how you ship a number that looks like growth and ship away the growth itself.

Critique 4 — "Proven at scale" inverts selection bias into a credential

The conclusion argues v250's large majority share proves the effect at scale. This inverts the logic: winning versions were ramped precisely because they looked good, so v250's share is the consequence of selection, not independent proof of its effect. The flat cancellation rate (0.315 → 0.311) is read as a clean win; on a platform whose biggest leak is cancellation, a flat rate means the intervention did nothing for the one thing that matters most, and a pooled rate can hide attribution gaming if drivers stall to force the rider to cancel. The recommendation then leaps to "start the Delhi/Kolkata rollout this sprint," with zero evidence from any city but Bangalore.

Ship implication: every safeguard that should slow a rollout (absent control group, flat cancellation, untested geography) is reframed as a reason to go faster. A process that treats its own selection bias as proof will keep shipping until something breaks.

What the dataset cannot tell us

The cube describes trips NY served. The answer to "why are rides flat" lives in the trips and people it lost. Five gaps matter, each with a way to close it:

- **Silent no-quote.** A rider who got no usable quote and left for another app is not in the funnel. Fix: log every search as quote-shown, no-quote, or quote-rejected, with the quoted fare attached.
- **No ids.** No retention, no repeat-rider cohort, no earnings-per-hour (the exact number the driver in Part 3 optimises). Fix: privacy-safe persistent ids.
- **No geography.** No pickup distance, no zone-level supply-demand mismatch; the Part 4 supply-shaping lever has no signal to run on without this. Fix: log pickup distance, tag zones.
- **No view of the driver's outside option.** Competitor rider fares are visible; competitor driver payouts are not. Fix: driver multi-homing panel and competitor incentive scrape.
- **No exogenous controls.** Weather, exam calendar, competitor subsidy events can swing rides for two weeks and land in the rollout window by coincidence. Fix: merge and subtract.

The deepest fix is the one the rollout never had: a real randomized holdout. Because assignment was observational and outcome-dependent, no amount of clever slicing fully separates the pricing from the confounds. A small held-back control, randomized within zone and hour, would answer in two weeks what this entire dataset cannot answer at all. It is the one thing I would insist on before any city-wide ship.

CLOSE

Summary for ops + exec

Bottom line. The congestion charge works in one narrow place and is getting credit for a lift it mostly did not cause; scaling it is the wrong move, and the real leak sits elsewhere.

Do. Keep the charge where it earns its place: Auto Priority at peak, held below the ~1.5x ceiling. Put the engineering behind the actual leak: ship a legible deadhead compensation offer card to cut driver cancellations, and start building supply-shaping, which buys completion with data NY already owns instead of relying on capital it does not have.

Stop / hold. Do not scale congestion pricing to the mass Auto tier or to 100%, and do not scale based on this evidence. The headline +14.9% is mostly contributed by a shrinking search base, demand that had already recovered before the charge shipped, and an off-peak version swap. The charge itself does almost nothing for the mass tier and nothing at all for cancellation.

One number to watch. Completed rides per day, the count and not any ratio, read Auto and Auto Priority apart. Before any city-wide ship, insist on a small randomized holdout: the one signal this entire dataset cannot stand in for.